

Alberta's Two Maps: What the Audit Found

A plain-language summary of the 2025–26 Electoral Boundary

Commission forensic audit. v0.19 — April 24, 2026 — [Full monograph](#)

An Airdrie resident in her thirties pulls up the new electoral map on her phone. She has lived in Airdrie for seven years. She knows her city — 84,000 people, larger than Red Deer, one city council, one property-tax bill. Under the majority commission proposal, Airdrie sits in two electoral divisions. Under the minority proposal, it is cut into four, each quarter attached to a different surrounding region. Her street goes to one riding. Two blocks over goes to another. Nothing about Airdrie changed. Only the map did.

That map is what this audit is about.

The short version

Alberta's Electoral Boundary Commission finished its work on March 23, 2026. It could not agree. The majority of commissioners produced one map. The minority produced a different one. Both maps are legal. This audit measured both using the same methods, applied identically to each. Three findings stand out.

1. **The two maps diverge systematically.** The minority map differs from the majority on six measurable dimensions: population spread, geographic concentration of voters, city fragmentation, municipal-boundary anchoring, map shape, and commission-chair-flagged anomalies. The differences are consistent across all six and none of the six require election results to calculate.
2. **The direction runs against the places where the governing party is weakest.** Communities most affected by the structural differences in the minority map — Calgary's northwest quadrant, the City of Airdrie, urban areas with well-established municipal boundaries — are the communities where the governing party's opponents are strongest. The audit cannot determine intent. It can measure effect.

3. **The process now promoting the minority map departs from Canadian practice.** On April 16, the government set both commission maps aside and assigned redistricting to a five-MLA committee, three from the governing UCP. Most Canadian provinces either require the assembly to consider the commission report first or give commission maps automatic effect unless overridden. Alberta does neither. The process is its own finding, separate from the maps.

Three things the minority map does differently

It splits the City of Airdrie into four pieces.

Airdrie is a city of about 84,000 people — larger than Red Deer. The law says each electoral division can hold at most one-and-a-quarter times the provincial average, so Airdrie needs at least two. The majority map puts Airdrie in two divisions. The minority map puts it in four — attaching each quarter to a different surrounding region. Both maps are legal. The four-way split is a choice.

It ignores municipal boundaries.

When electoral maps follow the edge of a city or town, voters recognize where their division begins and ends. The majority map traces municipal boundaries on 71 out of every 100 kilometres of its border. The minority map traces them on only 15 out of every 100 kilometres. This number has nothing to do with which party benefits — it is measured from publicly available maps, not from election results.

One area of Calgary is carved up in a way that concentrates NDP voters into larger-than-average divisions.

In the northwest quadrant of Calgary, the minority map's divisions are, on average, 11.5% larger than the province-wide average. The majority map's equivalent area is only 2.8% above average. When voters are concentrated into larger-than-normal divisions, their votes are less likely to contribute to electing anyone — a redistricting concept called "packing."

These three cases are illustrative. The audit found two more measures where the same pattern appears. In each case: both maps are legal; the minority map shows a larger departure from standard practice; the departure does not depend on vote data.

What the audit found across five measures

The table below compares the two maps. None of the top six rows use election results.

WHAT WAS MEASURED	MAJORITY MAP	MINORITY MAP
How evenly distributed the populations are	More even	48% more uneven
NW Calgary population excess above average	2.8%	11.5%
Airdrie split	2 divisions	4 divisions
Borders that follow existing municipal lines	71%	15%
Borders anchored to Statistics Canada census boundaries	80%	17%
Boundaries flagged by the commission chair	0	3
Efficiency gap (an election-fairness measure)	-1.3%	-2.7%
Packing-cracking neighbourhood pattern	3 (same as 2019)	0 — <i>exoneration</i>

The bottom two rows use election results and are less certain. The efficiency-gap numbers mean both maps favour the UCP relative to a perfectly neutral outcome — which is expected in Alberta because NDP voters are concentrated in cities, and this tends to produce a structural UCP advantage under any map. Both numbers are far below the threshold where researchers say a map looks like a deliberate gerrymander.

The last row is where the minority map actually looks better than the majority. One specific test — whether packed districts tend to sit next to cracked districts — found zero such pairs on the minority map, compared to three on the majority and three under the current 2019 map. The audit committed to this test before looking at the results, and when the minority map passed, it recorded the pass. That finding is called an **exoneration**.

One way to read the scorecard

Rows of measurements are hard to weigh against each other. One defensible approach is a four-category rubric — weights drawn from the principles that redistricting law and democratic theory say should matter most. This is a proposed synthesis, not a finding. Other weightings are possible; the individual measurements do not change.

CATEGORY	WEIGHT	WHAT IT MEASURES	MAJORITY	MINORITY
Voter equality	25%	Population spread; whether any area concentrates voters above the packing threshold	B	D
Community recognition	35%	How much the map follows city and town lines; how many pieces a single city is split into	B	F
Map coherence	25%	Whether district shapes have obvious explanations; chair-flagged anomalies	A	D
Partisan fairness	15%	Efficiency gap; packing signature; neighbour-drain exoneration	B	B
Under this rubric	—	—	B (87/100)	F (47/100)

Why these particular weights. Community recognition carries the most weight (35%) because the Saskatchewan Reference — the Supreme Court decision that defines what redistricting is for — holds that effective representation requires more than population parity: it requires districts that communities can identify as their own. The Act's $\pm 25\%$ population band grounds voter equality at 25%. Map coherence (25%) reflects the commission's own standard: the chair flagged specific boundaries as anomalous, and those flags matter. Partisan fairness is weighted lowest (15%) because both maps fall below any published gerrymander threshold, and the direction of the finding reverses depending on which measurement method you use. A reader who weighted partisan fairness more heavily would get different grades; the measurements supporting each category are in the table above this one.

What produces the minority map's F under this rubric. Both maps are legal. The F is not about intent — it reflects what the map produces for voters on the ground. On community recognition (10/35 under these weights), 14.5% municipal-boundary anchoring means 85 out of every 100 kilometres of the minority map's border cuts through territory voters do not recognize from any other civic map — not their school division boundary, not their property-tax bill, not their local election ward. The majority's equivalent figure is 71%. Splitting Airdrie's 84,000 residents into four districts each identified with a different surrounding region has a similar effect: Airdrie voters navigate four different legislative offices, each primarily identified with somewhere else. Three chair-flagged anomalies — a lasso-shaped corridor, a boundary extension through uninhabited national park land, and a district named for communities that together are smaller than the community it actually captures — pull the map-coherence score down further.

Where the minority map scores comparably. On partisan fairness, the two maps are close. The minority map produces zero packed-district / cracked-district adjacency pairs — a pattern present on both the 2019 map and the majority. (This finding is reported as a §5.3.5 exoneration in the full audit, committed before the results were read.) The majority map produces a slightly smaller efficiency-gap figure by one measurement method. Neither map has a serious partisan-fairness problem by any published standard. Under this rubric's lowest-weighted category, the two maps are effectively tied.

What the audit does not claim

This audit does not say the minority map was drawn to help the UCP win more seats. Proving intent is not possible from maps alone. What it does say is that the minority map differs from the majority in ways that are measurable, consistent, and not explained by Alberta's geography.

It also does not say the minority map is illegal. Both maps satisfy the *Electoral Boundaries Commission Act*.

The partisan-fairness numbers (the efficiency gap and related metrics) are sensitive to which election's results you use and how you measure them. Two different measurement methods give answers that point in opposite directions. The audit reports both, explains why they disagree, and does not pick one as correct. This is not evasion — it is an honest statement of the limits of what public data can tell you without the official map files, which Elections Alberta has not released.

The gerrymander checklist

Before looking at either map's results, the audit committed to a checklist for what a gerrymander would look like. This is called pre-registration — you write down what you expect to find before you look, so no one can accuse you of cherry-picking.

Here is the checklist, applied to both maps:

SIGNAL	MAJORITY MAP	MINORITY MAP
Packing signature (large divisions concentrated in one area)	Not detected	Detected (NW Calgary)
Cracking signature (close-loss divisions concentrated nearby)	Not detected	Partial — held pending more data
Geographic anomaly (lasso shape, extension through empty land)	Not detected	Detected (Nolan Hill–Cochrane lasso; RMH–Banff Park extension)
Community incorrectly claimed in justification	Not detected	Detected (school-division crossings, Cochrane commuter claim overstated)
Packing-cracking neighbourhood pairs	3 present	0 present — <i>exoneration</i>

Under the checklist's threshold ("three signals plus an outlier on the statistical ensemble"), the minority map comes close but does not clearly cross the line. It meets three of the signals, and it sits outside the 95th percentile on one statistical measure. It does not introduce new types of anomalies, and it does not clearly overturn the public submissions received by the commission.

The honest scorecard: the minority map has more anomalies than the majority, and those anomalies are consistent with a UCP-favouring pattern, but do not clearly meet the threshold for a definitive gerrymander finding.

The April 16 decision

The *Electoral Boundaries Commission Act* lets the government refer the commission's recommendations to a legislative committee. On April 16, the government did this. The new committee has five members — three from the governing UCP and one each from the NDP and the Alberta Party. The committee must report by November 2, 2026, with a 91-seat map.

The audit's procedural review found that this referral followed the letter of the Act. It also found that Alberta is an outlier among Canadian provinces in two ways: most provinces require the commission's report to be considered by the assembly before referral, and most provinces have commissions chaired by a judge rather than a member of the public who can be replaced mid-mandate. The audit documents three comparator cases (Quebec 1992, Ontario 1996, British Columbia 2008) where similar pivots occurred, for context.

What you can do

1. **Read the full findings.** The monograph ([report_academic.md](#)) contains every number, every source, and every caveat. If you find a factual error, the [retraction conditions](#) tell you exactly what evidence would change each conclusion.
2. **Ask your MLA what the committee's process will be.** Specifically: - Will the committee publish the evaluation criteria it uses *before* drawing a single line? - Will it release the range of maps it considered, not just the one it chose? - Will it publish any prompts or inputs given to AI tools?
3. **Request the official shapefiles.** Elections Alberta has not published the 2026 electoral division map files. Without these, some of this audit's calculations cannot be fully verified or corrected. A Freedom of Information and Protection of Privacy Act request would require their release.

4. **Share this report.** The full monograph is written for researchers and journalists. This report is written for everyone else. If you think the commission's process matters, share what you found [here](#).
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About this audit

This audit was produced by Will Conner, a computer science student at Mount Royal University in Calgary. It was conducted as an independent research project with no funding from any political party or advocacy organization.

The audit used publicly available data: Elections Alberta's 2023 Statement of Vote, Statistics Canada's 2021 Census, and the commission's 2026 final report. All code is published in this repository and can be run independently to reproduce every number.

An AI assistant (Claude, made by Anthropic) helped draft and check the text. All factual claims were verified by the author against primary sources. The audit declares this openly because the same standard it applies to the commission — transparency about process — applies to the audit itself.

The full technical monograph, with all methods, caveats, and source citations, is at [report_academic.md](#).